MAT 3007 - 2024 Spring
Assignment 3 Solution

Problem 1
First, write the canonical form of the linear program:

$$\begin{aligned} \text{minimize} \quad & -500x_1 - 250x_2 - 600x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 + x_4 = 240 \\ & 3x_1 + x_2 + 2x_3 + x_5 = 150 \\ & x_1 + 2x_2 + 4x_3 + x_6 = 180 \\ & x_1, x_2, ..., x_6 \geq 0 \end{aligned}$$

Problem 1: Iteration 1

B	-500	-250	-600	0	0	0	0
4	2	1	1	1	0	0	240
5	3	1	2	0	1	0	150
6	1	2	4	0	0	1	180

In the initial tableau, the basic set is$\{4, 5, 6\}$, and the basic solution is $(0, 0, 0, 240, 150, 180)$. The objective function value is 0. By Bland's rule, we choose the first column as the pivot column and bring index 1 in basis. Then take MRT, we choose $\theta^* = \min\{240/2, 150/3, 180/1\} = 50$ and the second row will be the pivot row, and the outgoing index is 5. Use pivot entry (the number in red color) to do row combination and change all other entries in column 1 to 0.

Problem 1: Iteration 2

B	0	-250/3	-800/3	0	500/3	0	25000
4	0	1/3	-1/3	1	-2/3	0	140
1	1	1/3	2/3	0	1/3	0	50
6	0	5/3	10/3	0	-1/3	1	130

In the second tableau, the basic set is$\{4, 1, 6\}$, and the basic solution is $(50, 0, 0, 140, 0, 130)$. The objective function value is -25000. By Bland's rule, we bring index 2 in. The pivot column is column 2. Then by MRT, we choose $\theta^* = \min\{720, 150, 78\} = 78$. The second row is the pivot row and the outgoing index is 6. Do a similar row transformation to change all other entries in column 2 to 0 and we will derive the following tableau.
In the third tableau, the basic set is$\{4, 1, 2\}$, and the basic solution is $(24, 78, 0, 57, 0, 0)$. The objective function value is -31500. We bring index 3 in. The pivot column is column 3. In this case, the third row should be the pivot row and the outgoing index should be 2. Conducting row transformation and updating the tableau we will arrive at iteration 4.

Problem 1: Iteration 3

B	0	0	-100	0	150	50	31500
4	0	0	-1/2	1/2	-3/10	-1/10	57
1	1	0	0	0	14/15	-1/5	24
2	0	1	2	0	-1/5	3/5	78

Problem 1: Iteration 4

B	0	50	0	0	140	80	35400
4	0	1/2	0	1	-7/10	-1/10	153
1	1	0	0	0	14/15	-1/5	24
3	0	1/2	1	0	-1/10	3/10	39

In the last tableau, the basic set is$\{4, 1, 3\}$, and the basic solution is $(24, 0, 39, 153, 0, 0)$. The objective function value is -35400. Since all the values of reduced cost are positive, we have arrived at the optimality.

Problem 2

For the original problem, we can write down the iteration 1 simplex tableau as follows

Problem 2: Iteration 1

B	-2	-3	1	12	0	0	0
5	-2	-9	1	9	1	0	0
6	1/3	1	-1/3	-2	0	1	0

By Bland's rule, we bring index 1 in. The pivot column is column 1. By checking the parameters in column 1, only the second row can be the pivot row, since $-2 < 0$. Thus the outgoing index is 6. Then update the tableau, we will derive iteration 2. And we can see that the current basic solution is feasible, but the variable with a negative reduced cost has a column with all negative entries. Therefore, the problem is unbounded.

Problem 2: Iteration 2

B	0	3	-1	0	0	6	0
5	0	-3	-1	-3	1	6	0
1	1	3	-1	-6	0	3	0

Problem 3

We first solve an auxiliary problem (Phase I).

$$\begin{array}{ll} \text{minimize} & x_6 + x_7 + x_8 \\ \text{s.t.} & x_1 + 3x_2 + 4x_4 + x_5 + x_6 = 2 \\ & x_1 + 2x_2 - 3x_4 + x_5 + x_7 = 2 \\ & -x_1 - 4x_2 + 3x_3 + x_8 = 1 \\ & x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8 \geq 0 \end{array}$$

Use simplex tableau to solve the auxiliary problem.

Problem 3: Phase I - iteration 1

B	-1	-1	-3	-1	-2	0	0	0	-5
6	1	3	0	4	1	1	0	0	2
7	1	2	0	-3	1	0	1	0	2
8	-1	-4	3	0	0	0	0	1	1

In the initial tableau, the basic set is$\{6, 7, 8\}$, and the basic solution is $(0, 0, 0, 0, 0, 2, 2, 1)$. The objective function value is 5. By Bland's rule, we choose the first column as the pivot column and bring index 1 in basis. Here since the test ratio of row 2 and row 3 are the same, we just choose the first row as the pivot row, and the outgoing index is 6. Then update the tableau.

Problem 3: Phase I - iteration 2

B	0	2	-3	3	-1	1	0	0	-3
1	1	3	0	4	1	1	0	0	2
7	0	-1	0	-7	0	-1	1	0	0
8	0	-1	3	4	1	1	0	1	3

Now in the iteration 2, the basic set is$\{1, 7, 8\}$, and the basic solution is $(2, 0, 0, 0, 0, 0, 0, 3)$. The objective function value is 3. By Bland's rule, we choose column 3 as the pivot column and bring index 3 in basis. And no need for MRT, we will choose the third row as the pivot row, and the outgoing index is 8. Then update the tableau.

Problem 3: Phase I - iteration 3

B	0	1	0	7	0	2	0	1	0
1	1	3	0	4	1	1	0	0	2
7	0	-1	0	-7	0	-1	1	0	0
3	0	-1/3	1	4/3	1/3	1/3	0	1/3	1

And from the result in iteration 3, we have arrived at the optimality for the auxiliary problem. And the optimal value of the auxiliary problem is zero, which means the original problem has

a feasible solution. Then we will start solving the original problem with the BFS provided by the above auxiliary problem. Since one of the slack variables (x_7) is still in the basis, and its optimal solution equals 0, we can just choose one variable from the rest to be the basic variable. Here, we choose x_2 to replace x_7. Next, use the simplex method to solve the original problem.

Problem 3: Phase II - iteration 1

B	0	0	0	3	-5	-7
1	1	0	0	-17	1	2
2	0	1	0	7	0	0
3	0	0	1	11/3	1/3	1

For the first iteration in phase II, we will choose column 5 as the pivot column, and MRT helps determine row 1 as the pivot row. Then we can update the tableau by conducting row transformation.

Problem 3: Phase II - iteration 2

B	5	0	0	-82	0	3
5	1	0	0	-17	1	2
2	0	1	0	7	0	0
3	-1/3	0	1	28/3	0	1/3

For the second iteration in phase II, we will choose column 4 as the pivot column, and MRT helps determine row 2 as the pivot row. Then we can update the tableau by conducting row transformation.

Problem 3: Phase II - iteration 3

B	5	17/7	0	0	0	3
5	1	17/7	0	0	1	2
4	7/4	17/4	0	1	0	0
3	-1/3	-4/3	1	0	0	1/3

From figure "Problem 3: Phase II", we can see the optimal solution is $(0, 0, \frac{1}{3}, 0, 2)$, and the optimal objective function value is -3.

Problem 4

(a). $\alpha \leq 0$, $\gamma \leq 0$, $\delta < 0$, $\beta \geq 0$. The current basic set is feasible. So, the first variable should enter the set. However, the corresponding column has entries either negative or zero.
(b). $\beta > 0$. The current solution is not optimal since the existence of negative reduced cost.

So, we only need to guarantee the feasibility. (Note that if $\beta = 0$, the problem case (c) may happen.)
(c). We want one of the basic solutions to be zero, so when it is added to the top row, the optimal value doesn't change. Also, we need to make sure that when making the reduced cost zero, the other reduced costs stay non-negative. There may be two sets of possible sufficient conditions for different cases.

Sufficient Conditions 1: $\beta = 0$, $\delta \geq 0$, $\delta \geq -\frac{2}{3}\gamma$.

For the first case, we need to ensure $\beta = 0$ first, so we may bring a non-basic variable in to substitute x_5 but do not change the optimal value. Then, since $\delta \geq 0$, column 2 will be the pivot column, and we are going to bring index 2 in and let index 5 out. Then calculate the updated reduced cost of nonbasic variable (x_1, x_5) in the next iteration, we want $\bar{c}_1 = \delta + \frac{2\gamma}{3} \geq 0, \bar{c}_5 = \frac{2}{3} \geq 0$, which lead to $\delta \geq -\frac{2}{3}\gamma$.

Sufficient Conditions 2: $\beta = 0$, $\delta \leq -\frac{2}{3}\gamma, \gamma > 0$. ($\delta < 0$)

Still $\beta = 0$ first. Then we can choose column 1 as the pivot column and let index 5 out, which requires $\gamma > 0$. Then calculate the updated reduced cost of nonbasic variable (x_2, x_5) in the next iteration, we want $\bar{c}_2 = -2 - \frac{3\delta}{\gamma} \geq 0, \bar{c}_5 = -\frac{\delta}{\gamma} \geq 0$, which lead to $\delta \leq -\frac{2}{3}\gamma$.

Problem 5
(a). $\beta \geq 0$.
(b). $\alpha \geq 0$, $\beta < 0$. The sum of all the positive variables has a negative value, which indicates infeasibility.
(c). $\beta \geq 0$, at least one of $\delta, \gamma, \xi < 0$. The reduced cost of one of the non-basic variables is negative.
(d). $\beta \geq 0$, $\alpha \leq 0$, $\delta < 0$. The fourth column has all entries negative or zero.
(e). $\beta \geq 0$, $\gamma < 0$, $\frac{2}{\eta} < \frac{3}{2}$, and $\eta > 0$. By the minimum ratio test, we want η to be the pivot element.